NATIONAL
Junior College

SH2 Preliminary Examination 2025

9173 H2 Geography Paper 1

Suggested Marking Guide

Disclaimer:
The Suggested Marking Guide should be treated with confidentiality, and should be used with discretion.

Notes on Marking

Marking Approaches

a. **Point marking** is used for marking all sub-parts of the structured questions, except for the 10-mark fieldwork evaluative sub-part.

   For point marking, the general rule is that each creditworthy response is awarded one mark. More detailed guidance is provided in the marking guide for each question.

b. **Generic level descriptors** are used for marking:

- 20-mark essays.

Possible approaches are provided for questions marked using generic level descriptors. They provide guidance on how the questions could be approached. The possible approaches provided are **neither exhaustive nor should they be treated as model approach to questions**.

Generic Level Descriptors

Levels	Marks	Generic Level Descriptors for H2 Essays
5	18–20	Evaluation is consistently analytical and coherent. Response is well-supported by relevant material, including the effective use of examples. Response features accurate geographical knowledge and reflects good understanding of the subject content relevant to the question.
4	14–17	Evaluation is analytical and coherent. Response is mostly well-supported by relevant material, including the appropriate use of examples. Response features accurate geographical knowledge and reflects adequate understanding of the subject content relevant to the question.
3	10–13	Evaluation is broadly analytical and generally coherent. Response is moderately well-supported by relevant material, including some appropriate use of examples. Response features accurate geographical knowledge and reflects adequate understanding of the subject content relevant to the question.
2	6–9	Response is largely descriptive with limited analysis and evaluation. Response is partly coherent and may lack clarity in parts. Response is poorly supported by relevant materials, including the limited use of examples. Response features inaccurate geographical knowledge and poor understanding of the subject content relevant to the question.
1	1–5	Response is descriptive with no analysis or evaluation. Response is fragmented and lacks clarity. Response consists of unsupported assertions. Response features largely inaccurate geographical knowledge and a lack of understanding of the subject content relevant to the question.
0	0	No creditworthy response

Note:
Assessment using generic level descriptors involves qualitative rather than quantitative evaluation. Judgements on the level to be awarded to an answer will be based on the principle of 'best fit' determined by the descriptors within each level.

As a general guideline, responses deemed to have fulfilled all the descriptors within a level may be awarded the top mark in that level.

Section A

Cluster 1: Development, Economy and Environment

Question	Answer	Marks
	Resource 1 shows the global energy consumption by source. Resource 2 shows the hydropower projects in the world in 2020. Resource 3 shows a hydropower dam in China. Resource 4 shows China's installed generation capacity of different energy sources and its energy progress report towards achieving Sustainable Development Goal (SDG) 7: Affordable and Clean Energy.	
1(a)	**Using an example for each, explain what is meant by renewable and non-renewable resources.** *Award 1 mark for each explanation using an example of what is meant by renewable or non-renewable resources.* *Award a maximum of 1 additional mark for further development of each explanation.* *Award a maximum of 2 marks for explanation of <u>either</u> renewable <u>or</u> non-renewable resources only.* Possible responses include: • Hydropower is a **renewable resource** because it **does not run out** and **can be replenished naturally** at a rate equal to or faster than it is consumed. o It generates electricity by using the **energy of moving water**—usually from rivers or dams. It uses the **natural water cycle**, which is powered by the sun. • Coal is a **non-renewable resource** because it is **finite,** and **once used, cannot be replaced** at a rate equal to or faster than it is consumed. o Because of its non-renewable nature, there are coal mines that have been depleted and abandoned. **AO1**	**4**
1 (b)	**Cite data from Resource 1 to compare the actual and projected global energy consumption in 2010 and 2050.** Award 1 mark for each comparison with the use of data, up to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each comparison, where applicable. Possible responses include:	**4**

Question	Answer	Marks
	• The total actual global energy consumption will **increase substantially** from **600** quadrillion BTU in **2010** to a projected **900** quadrillion BTU in **2050**. • **Petroleum and other liquids** used to be the **top global energy source**, accounting for **1/3 of the total consumption in 2010** but is projected to **tie with renewables** by being **joint 1st in 2050**. o Petroleum and other liquids accounted for **200** quadrillion BTU in **2010** to a projected **250** quadrillion BTU in **2050**. • The consumption of **natural gas** is expected to **increase gradually from 160 to 180** quadrillion BTU in **2050**. **AO2**	
1 (c)	**Suggest reasons for the trends described in (b).** *Award 1 mark for each reason, up to a maximum of 6 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable.* Possible responses include: The projected increase in total global energy consumption • **Global population growth** from 2010 to 2050: More people mean greater demand for energy in transport, housing and industry. • **Economic development**, especially in emerging economies (e.g., India, Southeast Asia, sub-Saharan Africa), **increases energy needs** for manufacturing, services, and consumer lifestyles. Losing dominance in petroleum and other liquids though some demand continues • **Continued demand** in sectors like **transport and petrochemicals**, especially in countries with limited infrastructure for other sources of energy such as hydropower and solar power. • **Climate policies and carbon reduction targets** are discouraging investment in fossil fuels, especially in power generation. Increasing dominance in renewables • **Falling costs** of renewable technologies like solar, wind and battery storage make them more economically viable. • **Government incentives** and investments in clean energy infrastructure. **AO2**	**6**

Question	Answer	Marks
1 (d)	**Describe the spatial distribution of the hydropower projects in the world shown in Resource 2.** *Award 1 mark for each description, up to a maximum of 5 marks.* *Award a maximum of 1 additional mark for further development of each description, where applicable.* Possible responses include: • Hydropower projects are **not evenly distributed** across the world. • Most projects are **concentrated in developing regions,** with the three largest regions being South Asia, East Asia, and Southeast Asia and Oceania. o The **largest concentration** of planned and under-construction projects is found in **South Asia with 118 GW, with more than 80% still in the permitting and development phase.** • **The hydropower projects in other regions are mostly still in the permitting and development phase,** except for **East Asia**, with **2/3 (66%) of the projects under construction.** • There are **fewer hydropower projects in the developed regions of Europe (8 GW)** and **United States and Canada (4 GW)**. **AO2**	**5**
1 (e)	**With reference to Resource 3, explain the social, economic and environmental trade-offs of using hydropower to achieve energy security.** *Award 1 mark for each explanation.* *Award a maximum of 1 additional mark for further development of each explanation, where applicable.* *Award a maximum of 2 marks if only one type of trade-offs is explained.* Possible responses include: • Social trade-offs: Hydropower can **improve living standards** due to **access to electricity** improving people's quality of life, especially in rural or underserved regions**.** • However, it may also lead to **social disruption, inequality and cultural loss** if poorly planned. • Economic trade-offs: Hydropower provides **low-cost, stable energy over the long term** once built and **reduces reliance on imported fossil fuels**, improving energy independence. • However, it requires **significant investment** and returns depend on **climate reliability and social acceptance**.	**6**

Question	Answer	Marks
	• Environmental trade-offs: Hydropower and dams help with **flood control** and water storage. Reservoirs can support **irrigation** and **recreation**. • However, they can cause **ecological damage and loss of biodiversity** due to habitat destruction and changes in river flow. **AO1**	
1 (f)	**With reference to Resource 4, describe the challenges associated with China's attainment of SDG 7 by 2030.** *Award 1 mark for each description.* *Award 1 additional mark for further development of each description, where applicable.* Possible responses include: • One challenge is that **coal dominates China's energy mix**, accounting for **38% of installed capacity**. • Coal is a **major source of carbon emissions and air pollution** and hence limits progress toward clean and sustainable energy. • Shifting away from coal will be politically and economically difficult due to **industrial dependence and job implications**. • Transitioning to **cleaner sources** (like solar, wind, or hydro) demands massive **infrastructure upgrades and financial investment.** • Only **15% of total energy consumption comes from renewables in 2023**, **despite** China having **high solar (26%) and wind (16%) *installed capacity***. **AO2**	**5**

Cluster 2: Tropical Environments

Question	Answer	Marks
2(a)	**With reference to Resources 5 and 6, account for the relationship between sediment size and downstream distance in the Ganga River.** *Award 1 mark for each explanation of relationship between sediment size and downstream distance.* *Award a maximum of 1 additional mark for further development of each explanation where applicable.* Possible responses include: • **Sediment size decreases with increasing downstream distance**, showing finer particle size towards the lower course. o In the upper Ganga Plain, coarse particles like gravel and coarse sand dominate. By 400km downstream, fine sand dominates. By 2100km downstream, clay and silt dominate the bed load • **River velocity affects river competence** – the river's ability to transport particles or sediment of various weights and sizes. o Based on the **Hjulstrom Curve**, it is more difficult to transport coarser load due higher settling velocity, so they are confined upstream unlike the finer ones **AO2**	**4**
2(b)	**Using Resource 7, describe how the fluvial landforms of the Ganga River have changed between 1900 and 1963.** *Award 1 mark for each explanation of the change in fluvial landform up to a maximum of 3 marks.* *Award a maximum of 1 additional mark for further development of each explanation, where applicable.* Possible responses include: • Resource 7 shows the Ganga River become **increasingly sinuous and wider** near Munger between 1900 and 1963. • In 1900, the channel was **relatively narrow and at the outer bank, closer to the river terrace margin** • **More braided sections appeared** within the active channel area in 1936 • By 1963, the river had **become wider and more sinuous near Munger.**	**5**

Question	Answer	Marks
	o Several mid-channel bars have consolidated into a much larger single deposition of sediment close to the river valley terrace **AO2**	
2 (c)	**With reference to Resources 5, 6 and 7, suggest factors that could influence changes in fluvial landforms of the Ganga River over time.** *Award 1 mark for each suggestion of factors up to a maximum of 6 marks.* *Award 1 mark for further development of each explanation, where applicable.* Possible responses include: • Topography - Resource 5 shows the Ganga traverses through the **alluvial Ganga and Bengal plains from the Himalayas**. These are **floodplain zones with high sediment availability.** • **The river terraces** shown in Resource 7 are evidence of past floodplain levels when the Ganga plains might have carried a high sediment load during wet monsoon seasons. In the **development of meanders** shown in Resource 7, river terraces may also **limit the lateral erosion** of the Ganga River and **promote vertical erosion** instead. • Factors at the Channel Level - Resource 6 indicates the downstream load of sediments including coarse sand and gravel to silt and clay. This reflects **long transport distances, selective deposition and diminishing stream competence** • Resource 7 reveals **significantly increased channel braiding and bar formation** typical of river systems that are **over-loaded with sediment** and high river capacities. The load may also increase the amount of friction that may reduce river velocities and result in deposition • Climate - **Monsoonal rainfall** in tropical environments as shown by the location of Ganga Plain at about 23-25◦ latitude as shown in Resource 5, may take place across the drainage basin and **contribute to high seasonal discharge** • **Flashy hydrographs** during high-intensity monsoon rainfall promote bank erosion. **Himalayan snowmelt** may also increase the propensity of **slope wash** occurring, and increase sediment yield in tropical humid tropics	**6**

Question	Answer	Marks
	AO2	
2 (d)	**Using Resource 8, explain how projected changes in the Ganges-Brahmaputra delta from 2020 to 2080 could cause fluvial floods.** *Award 1 mark for each explanation of change that may cause fluvial flood to occur up to a maximum of 5 marks.* *Award 1 mark for further development of each explanation, where applicable.* Possible responses include: • **Significantly increased urbanisation from 17.9% in 2020 to 30.2% in 2080 may dramatically raise levels of surface runoff.** • **Urbanisation has mostly taken place around river distributaries**. Urban settlements **increase impervious surfaces** (e.g. roads, buildings) and this significantly raises runoff coefficients. • **Reduction of cropland** - Resource 4 shows cropland potentially decreasing from **55.6% from 2020 to 43.2% in 2080**. This reduces interception and infiltration capacity. • **Deforestation in upper and middle delta zones** may occur with potential reduction of 0.1%. This reduces root cohesion and slope stability, increasing soil erosion. • **Small increase of wetland areas by 0.5% from 2020 to 2080** may increase the delta's ability to store and slowly release floodwaters, although this may occur only marginally. **AO2**	**5**
2 (e)	**With reference to Resource 8, explain how human activities in the Ganga River basin could have an impact on ecosystem services.** *Award 1 mark for each explanation of impact on ecosystem service up to a maximum of 4 marks.* *Award 1 mark for further development of each explanation, where applicable up to a maximum of 2 marks.* Possible responses include: • **Provisioning services** (E.g., fertile alluvial soils) are undermined by excessive sedimentation and nutrient depletion downstream of dams. Sand mining, dam construction, and land reclamation **interrupt natural sediment transport, harming deltaic ecosystems**.	**6**

Question	Answer	Marks
	• **Regulating services** such as flood moderation and water purification decline with the destruction of riparian vegetation and wetlands. • **Supporting services** like biodiversity and nutrient cycling are impaired by channel engineering, altered hydrology, and siltation of aquatic habitats. These changes **reduce resilience and increase socio-ecological vulnerability** in a densely populated basin. • **Cultural services** including spiritual sites along the Ganga are disrupted by pollution and altered flow regimes. **AO1**	
2 (f)	**Suggest the strengths and limitations of two strategies to manage fluvial floods.** *Award 1 mark for each explanation of strength and limitation for each strategy, up to a maximum of 4 marks.* *Award a maximum of 2 marks if only strengths or only limitations are addressed.* Strategy: **Construction of levees/embankments along the river and distributaries**. • Strength: Provides **immediate protection to settlements** from annual monsoonal floods; relatively low-tech and can be implemented quickly. • Limitation: Embankments **constrain natural floodplain processes** and, when overtopped, may cause catastrophic breaches; maintenance-intensive and often poorly funded. **Strategy: Construction of Dams to provide temporary storage of water so that downstream flood peaks are lowered.** • Strength: Dams and reservoirs help **regulate river discharge by storing excess water** during heavy rainfall thereby reducing downstream flood peaks. • Limitation: Dams in tropical environments are prone to rapid sedimentation due to **high sediment loads** which significantly **reduces their storage capacity over time**. **AO1**	**4**

Section B

Cluster 1 Development, Economy and Environment

Question	Answer	Marks
3	**Evaluate the usefulness of Malthus' theory in understanding current global challenges related to population growth and provisioning services.** Possible Approaches: Candidates could approach the question by making a judgement on the usefulness of Malthus' theory through **assessing the relevance and limitations** of Malthusian thinking when applied to **contemporary global challenges**, such as food insecurity, water scarcity, climate change, rapid urbanisation and resource depletion. Candidates could also approach the question by making a judgment on whether Malthus' theory is useful in today's context through a consideration of two or more case studies. Candidates may argue that Malthus' theory still provides useful insights in contexts of **unsustainable population growth**, **environmental degradation** and **unequal access to resources**. Candidates may argue that Malthus did not consider **demographic shifts** which happen in tandem with economic development and that **Boserup's theory**, showing optimism about **technological innovation,** is a more useful framework in today's world.. *Levels marked using H2 generic level descriptors for 20m H2 essays.* **AO3**	**20**

Question	Answer	Marks
4	**Evaluate the influence of multilateral institutions on economic activities.** Possible Approaches: Candidates could approach the question by evaluating the extent to which multilateral institutions influence economic activities **directly** (e.g. through conditional loans or trade arbitration) and **indirectly** (e.g. through shaping policy environments, investor confidence, or integration into global markets). Candidates could evaluate both **positive and negative impacts**, considering how these institutions promote economic development, structural reform and financial stability; influence trade rules, financial flows, investment and policy	**20**

	decisions, and may also reinforce inequality and dependency in some regions. Candidates could approach the question by making a judgment on through a consideration of two or more case studies **of multilateral institutions (e.g. the World Bank, WTO and ASEAN) in shaping economic activities** across countries and sectors. Candidates could consider **varied impacts across different contexts**, such as **CHLDs and CLLDs**, **export-led vs. resource-dependent economies** and **countries with strong vs. weak governance**. *Levels marked using H2 generic level descriptors for 20m H2 essays.* **AO3**	

Section C

Cluster 2 Tropical Environments

Question	Answer	Marks
5	**'The Hadley cells are the main influence on rainfall patterns in the tropics.'** **Evaluate this statement.** Candidates could approach the question by arguing that while the Hadley Cells play a **foundational role in setting global circulation patterns**, the synoptic scale circulation significantly modify rainfall patterns across different parts of the tropics. The Hadley Cells provide a **global framework**, but **local and regional factors may modify their influence seasonally or interannually**, such as the synoptic scale circulation of **monsoons circulations** and **ENSO,** and discuss **seasonal and interannual variation** in rainfall in the tropics. Candidates could use **specific tropical case studies** (e.g. equatorial Africa, the Amazon, Southeast Asia, or South Asia) to illustrate rainfall variability in terms of **spatial and temporal differences** within the tropics. Candidates could evaluate the relative significance of the different factors in each case study and analyse the **interplay between the factors**. *Levels marked using H2 generic level descriptors for 20m H2 essays.* **AO3**	**20**

Question	Answer	Marks
6	**'Geology is the most important factor influencing the formation of karst landforms in the humid tropics.'** **Evaluate this statement.** Candidates could approach the question by explaining how **geology**, particularly the **type, rock composition and structure, purity, and jointing of carbonate rocks** (e.g. limestone or dolomite), provides the fundamental basis for karst development by influencing processes of **carbonation and solution**. Candidates could evaluate other factors, e.g. climatic, hydrological, biological and human factors, that play a significant role in the rate, extent, and variety of landform development in the humid tropics, e.g. **high rainfall and warm temperatures** enhance chemical weathering and intensify karstification. Candidates could argue that **geology is a necessary but not sufficient condition**—karst cannot form without soluble rock, but its characteristics and expression are **highly dependent on climate, hydrology, and biological factors**, especially in the tropics. **Climatic factors in the humid tropics** (e.g. abundant rainfall, biological productivity) may **accelerate karstification more than geological structure alone**. *Levels marked using H2 generic level descriptors for 20m H2 essays.* **AO3**	**20**